1. Explain the Types of Comments in Python.

Comments are used to explain the code and make it easier to understand. Comments are ignored by the Python interpreter during program execution.

There are two common ways to write comments in Python:

a) Single-Line Comment: A single-line comment begins with the # symbol. It is used to write a comment on one line.

Example:

```
# This is a single-line comment

print("Hello World")
```

b) Multi-Line Comment: A multi-line comment is used to write explanations across multiple lines. We can use the # symbol on each line or use triple-quoted strings (''' or """) for documentation, although triple-quoted strings are not technically comments.

Example:

```
# This is a comment

# written on multiple lines

print("Hello World")
```

2. What Are the Two Ways to Execute a Python Program?

There are two ways to execute a Python program:

a) Interactive Mode: In interactive mode, we type Python commands directly into the Python interpreter and get the result immediately. It is useful for testing small pieces of code.

Example:

```
>>> print("Hello World")

Hello World
```

b) Script Mode: In script mode, we write Python code in a file with the .py extension and execute the file using the Python interpreter. It is useful for writing and running complete programs.

Example:

Create a file named hello.py:

```
print("Hello World")
```

Run the file using:

```
python hello.py
```

Output:

```
Hello World
```

3. What Is a Multi-Line Statement?

A multi-line statement is a statement written across two or more lines. In Python, long statements can be continued on the next line using a backslash (\) or by enclosing the expression in parentheses, brackets, or braces.

Example 1: Using a backslash

```
total = 10 + 20 + \
    30 + 40
print(total)
```

Output:

100

Example 2: Using parentheses

```
total = (10 + 20 +
     30 + 40)
print(total)
```

Output:

100

Thus, multi-line statements allow long expressions to be divided across multiple lines, making the code easier to read.